DO NOW — RECALL

Lesson 080 · Insertion Sort: Grow the Sorted Prefix

Name: _____ Class: _____ Date: _____

1. What gets fixed after a bubble pass?

2. What does insertion-sort key mean?

3. What does sorted prefix mean?

4. BST root10 with right child15: insert key12. Show comparisons and position.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Largest remaining value at end.

2. Model answer

Item currently being placed.

3. Model answer

Initial already-ordered portion.

4. Model answer

12>10 so go right;12<15 so insert as left child of15.

By the end, you can…

• The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

• Shift larger elements one position to the right, then insert key in the vacant position.

• Worst $O(n^2)$, already-sorted best O(n), auxiliary O(1); especially useful on small/nearly sorted arrays.

Vocabulary — know these terms

insertion sort — Insert key into ordered prefix.

key — Value currently being inserted.

shift — Move existing value to make room.

sorted prefix — Already-sorted left section.

NEW LEARNING

Read, annotate and explain

Understand

The left portion is kept sorted. Read the next item as key and compare backwards with earlier elements.

Trace

Shift larger elements one position to the right, then insert key in the vacant position.

Explain

Worst $O(n^2)$, already-sorted best $O(n)$, auxiliary $O(1)$; especially useful on small/nearly sorted arrays.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insertion-sort [4,1,3]; give states after each key.

Check the working

Key1→[1,4,3]; key3→[1,3,4].

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give arrays after first two key insertions into [5,2,4,1]. [6]

Question 2

Complete insertion of final key1 into [2,4,5,1], showing working. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare insertion and bubble sort method/time on already sorted input. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Insertion sort always requires a separate second array to store sorted values.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.